

More: Sequences, Series, and Power Series FRQ

The Taylor series for a function f about $x = 1$ is given by $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{4^n} (x-1)^n$ and converges to $f(x)$ on its interval of convergence.

- (a) Using the ratio test, find the interval of convergence of the Taylor series for f about $x = 1$. Justify your answer.
- (b) Write the first three nonzero terms and the general term of the Taylor series for f' , the derivative of f , about $x = 1$.
- (c) Let $P_2(x)$ be the second-degree Taylor polynomial for f about $x = 1$. It is known that $0.1 \leq |f'''(x)| \leq 0.3$ on the interval $1 \leq x \leq 1.5$. Use Lagrange error bound to show that $P_2(1.5)$ differs from $f(1.5)$ by at most $\frac{1}{100}$.

The Taylor series for a function f about $x = 7$ is given by $\sum_{n=1}^{\infty} \frac{(x-7)^n}{n \cdot 5^n}$ and converges to $f(x)$ for all values of x in the interval of convergence.

- (a) Find the interval of convergence of the Taylor series for f about $x = 7$.
- (b) Write the first four nonzero terms of the Taylor series for f' , the derivative of f , about $x = 7$.
- (c) Let $g(x) = 2e^{f(x)}$. Find $g'(7)$. Show the work that leads to your answer.

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{(4)}(x)$
0	4	5	-1	$-\frac{15}{2}$	23
1	8	3	-2	$\frac{3}{2}$	$\frac{2}{5}$

Let f be a function having derivatives of all orders for all real numbers. Selected values of f and its first four derivatives are shown in the table above.

- (a) Write the second-degree Taylor polynomial for f about $x = 0$ and use it to approximate $f(0.2)$.
- (b) Let g be a function such that $g(x) = f(x^3)$. Write the fifth-degree Taylor polynomial for g' , the derivative of g , about $x = 0$.
- (c) Write the third-degree Taylor polynomial for f about $x = 1$.
- (d) It is known that $|f^{(4)}(x)| \leq 300$ for $0 \leq x \leq 1.125$. The third-degree Taylor polynomial for f about $x = 1$, found in part (c), is used to approximate $f(1.1)$. Use the Lagrange error bound along with the information about $f^{(4)}(x)$ to find an upper bound on the error of the approximation.

Let f be the function defined by $f(x) = \frac{1}{x^2 + 9}$.

(a) Evaluate the improper integral $\int_3^\infty f(x) \, dx$, or show that the integral diverges.

(b) Determine whether the series $\sum_{n=3}^\infty f(n)$ converges or diverges. State the conditions of the test used for determining convergence or divergence.

(c) Determine whether the series $\sum_{n=1}^\infty \frac{(-1)^n}{(e^n \cdot f(n))} = \sum_{n=1}^\infty \frac{(-1)^n(n^2 + 9)}{e^n}$ converges absolutely, converges conditionally, or diverges.

The Maclaurin series for a function f is given by $\frac{x}{3} + \frac{x^2}{4} + \frac{x^3}{5} + \cdots + \frac{x^n}{n+2} + \cdots$.

- (a) Use the ratio test to find the interval of convergence of the Maclaurin series for f .
- (b) Let g be the function given by $g(x) = f(-2x)$. Find the first three terms and the general term of the Maclaurin series for g .
- (c) The first two terms of the Maclaurin series for f are used to approximate $f(0.1)$. Given that $|f'''(x)| \leq 2$ for $0 \leq x \leq 0.1$, use the Lagrange error bound to show that this approximation differs from $f(0.1)$ by at most $\frac{1}{3000}$.

The Maclaurin series for a function f is given by $f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + \frac{(-1)^n x^{2n+1}}{(2n+1)!} + \cdots$ and converges to $f(x)$ for all x . Let g be the function defined by $g(x) = f(x^2)$.

- (a) Find the first four nonzero terms and the general term of the Maclaurin series for g .
- (b) Find $\lim_{x \rightarrow 0} \frac{x^2 - \frac{x^6}{6} - g(x)}{x^{10}}$.
- (c) Let h be the function defined by $h(x) = \int_0^x g(t) dt$. Write the first four nonzero terms and the general term of the Maclaurin series for h . Use the first two terms of your answer to approximate $h(1)$.
- (d) Explain why the approximation found in part (c) differs from the actual value of $h(1)$ by less than $\frac{1}{1000}$.